Class 17 Learning Activity

Matlab Work-Up

Before summarizing the assignment, the file, `activity_17.m`, was created for performing the calculations. Within it a function of the same name was declared. Each of the blocks of code listed below are added to this file to complete the function for performing the calculations.

```
function activity_17()
% Calculations for the Class 17 Learning Activity in REB, The Course
```

Assignment Summary

Reaction

$$2A \rightarrow 2Y + Z \tag{1}$$

Rate Expression

$$r_1 = k_1 C_A^2 \tag{2}$$

Reactor: Adiabatic, gas-phase BSTR

Given Constants: $k_{0,1}$ = 265 L mol^{-1} min^{-1}, E_1 = 73 kJ mol^{-1}, $\Delta H_1\big|_{298K}$ = -165 kJ mol^{-1}, P_0 = 1000 Torr, T_0 = 1225 K, and T_f = [10, 100] K

```
    % global constants available to all functions in consistent units
    % given
    k01 = 265; % L /mol /min
    E1 = 73000; % J /mol
    dH298 = -165000; % J /mol
    P0 = 1000; % Torr
    T0 = 1225; % K
    V = 1; % basis
```

```
Tf = [1235,1325]; % K
% known
Re = 8.314; % J /mol /K
Rw = 62.36367; % L Torr /mol \K
% calculated
nA0 = P0*V/(Rw*T0);
```

Given Equations

$$\hat{C}_{p,A} = 28 + 0.05T \tag{3}$$

$$\hat{C}_{p,Y} = 26 + 0.01T \tag{4}$$

$$\hat{C}_{p,Z} = 30 + 0.005T \tag{5}$$

```
% given equations
Cp_A = @(T) 28 + 0.05*T;
Cp_Y = @(T) 26 + 0.01*T;
Cp_Z = @(T) 30 + 0.005*T;
```

Task: Parametric analysis

- Parameter: T_f
- Quantities of Interest: t_f and $f_{A,f}$

Preliminary Analysis

The reactor is adiabatic and at 298 K the reaction is exothermic. Assuming it remains exothermic at reaction temperature, the temperature is expected to rise as the reaction occurs. There is only one reaction taking place, so the amount of A is expected to decrease and the amounts of Y and Z, to increase. The times may be small given the very high reaction temperature, but it will depend upon the rate coefficient.

BSTR Model Function

BSTR Design Equations

$$\frac{dn_A}{dt} = -2Vr_1 \tag{6}$$

$$\frac{dn_Y}{dt} = 2Vr_1 \tag{7}$$

$$\frac{dn_Z}{dt} = Vr_1 \tag{8}$$

$$\left(n_A\hat{C}_{p,A} + n_Y\hat{C}_{p,Y} + n_Z\hat{C}_{p,Z}\right)\frac{dT}{dt} - V\frac{dP}{dt} = -Vr_1\Delta H_1 \tag{9}$$

$$RT\left(\frac{dn_A}{dt} + \frac{dn_Y}{dt} + \frac{dn_Z}{dt}\right) + \left(n_A + n_Y + n_Z\right)R\frac{dT}{dt} - V\frac{dP}{dt} = 0 \tag{10}$$

BSTR Model Variables: $\underline{t}$, $\underline{n}_A$, $\underline{n}_Y$, $\underline{n}_Z$, $\underline{T}$, and $\underline{P}$

Additional Computable Unknowns: r_1, k_1, C_A, $\hat{C}_{p,A}$, $\hat{C}_{p,Y}$, $\hat{C}_{p,Z}$, and ΔH_1

$$k_1 = k_{0,1}\exp\left(\frac{-E_1}{RT}\right) \tag{11}$$

$$C_A = \frac{n_A}{V} \tag{12}$$

$$\Delta H_1^0 = \Delta H_1^0\Big|_{T=298K} + \int_{298\text{ K}}^{T}\left(-2\hat{C}_{p,A} + 2\hat{C}_{p,Y} + \hat{C}_{p,Z}\right)dT \tag{13}$$

Uncomputable Unknowns (Received as Arguments): T_f

IVODE Solver Requirements

1. Initial values

$$t_{iv} = 0$$

$$n_{A,iv} = n_{A,0} = \frac{P_0 V}{R T_0}$$

$$n_{Y,iv} = n_{Z,iv} = 0$$

$$T_{iv} = T_0$$

$$P_{iv} = P_0$$

2. Stopping criterion

$$T = T_f$$

3. BSTR derivatives function (see below)

```
% BSTR reactor function
function [t, nA, nY, nZ, T, P] = bstr_model_variables(Tf)
    % define the initial values
    ind_0 = 0;
    dep_0 = [nA0; 0; 0; T0; P0];

    % define the stopping criterion
    stopVar = 4;
    stopVal = Tf;

    % solve the design equations
    odes_are_stiff = false;
    [t, dep, success, message] = solve_ivodes(ind_0, dep_0, stopVar...
        , stopVal, @bstr_derivatives, odes_are_stiff);

    % check for solver issues
    if ~success
        disp('')
        disp(["BSTR model function issue: ", message])
        disp('')
    end

    % return the individual dependent variables
    nA = dep(:,1);
    nY = dep(:,2);
    nZ = dep(:,3);
    T = dep(:,4);
```

```
    P = dep(:,5);
end
```

BSTR Derivatives Function

1. receives the BSTR model variables at the start of an integration step
2. calculates additional unknowns using equations (2), (11), (12), and (13)
3. evaluates and returns design equation (residuals/derivatives) using equations (14) through (16).

$$\underline{\underline{M}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \sum_i \left(n_i \hat{C}_{p,i}\right) & -V \\ RT & RT & RT & R\sum_i n_i & -V \end{bmatrix} \tag{14}$$

$$\underline{g} = \begin{bmatrix} -2Vr_1 \\ 2Vr_1 \\ Vr_1 \\ -Vr_1\Delta H_1 \\ 0 \end{bmatrix} \tag{15}$$

$$\begin{bmatrix} \frac{dn_A}{dt} \\ \frac{dn_Y}{dt} \\ \frac{dn_Z}{dt} \\ \frac{dT}{dt} \\ \frac{dP}{dt} \end{bmatrix} = \underline{\underline{M}}^{-1}\underline{g} \tag{16}$$

```
% BSTR derivatives function
function ddt = bstr_derivatives(~, dep)
    % extract the dependent variables
    nA = dep(1);
    nY = dep(2);
    nZ = dep(3);
    T = dep(4);
```

```
        % calculate the additional unknowns
        dH_1 = dH298 -2*integral(Cp_A,298,T) + 2*integral(Cp_Y,298,T)...
            + integral(Cp_Z,298,T);
        r_1 = k01*exp(-E1/Re/T)*(nA/V)^2;

        % Create mass matrix, setting all elements to zero
        mass_matrix = zeros(5,5);

        % Add 1 on the diagonal for the first 3 rows
        mass_matrix(1,1) = 1.0;
        mass_matrix(2,2) = 1.0;
        mass_matrix(3,3) = 1.0;

        % Add the elements for the energy balance
        mass_matrix(4,4) = nA*Cp_A(T) + nY*Cp_Y(T) + nZ*Cp_Z(T);
        mass_matrix(4,5) = -V*Re/Rw;

        % Add the elements for the ideal gas law equation
        mass_matrix(5,1) = Rw*T;
        mass_matrix(5,2) = Rw*T;
        mass_matrix(5,3) = Rw*T;
        mass_matrix(5,4) = Rw*(nA + nY + nZ);
        mass_matrix(5,5) = -V;

        % Create right side vector
        rhs1 = -2*V*r_1;
        rhs2 = 2*V*r_1;
        rhs3 = V*r_1;
        rhs4 = -V*r_1*dH_1;
        rhs5 = 0.0;
        rhs = [rhs1; rhs2; rhs3; rhs4; rhs5];

        % Evaluate the derivatives to be returned
        ddt = mass_matrix\rhs;
    end
```

Items of Interest Function

1. For each specified value of T_f

a. Call the BSTR model function with that value as the argument to get $\underline{t}$, $\underline{n}_A$, $\underline{n}_Y$, $\underline{n}_Z$, $\underline{T}$, and $\underline{P}$

b. Extract the final time and calculate the final conversion.

$$t_f = \underline{t}\big|_{T=T_f} \tag{17}$$

$$f_{A,f} = \frac{n_{A,0} - \underline{n}_A\big|_{T=T_f}}{n_{A,0}} \tag{18}$$

2. Tabulate, show, and save the results.

```
    % items of interest function
    function items_of_interest()
        % allocate storage for the deliverables
        tf = [nan, nan];
        fA = [nan, nan];

        % loop through the Tf values
        for n=1:2
            % solve the bstr design equations
            [t, nA, ~, ~, ~, ~] = bstr_model_variables(Tf(n));

            % save the final time and the conversion
            tf(n) = t(end);
            fA(n) = 100*(nA0 - nA(end))/nA0;
        end

        % tabulate, show, and save the results
        item = ["Final T (K)"; "Time (min)"; "Conversion (%)"];
        value = [Tf; tf; fA];
        resultsTable = table(item, value);
        disp('')
        disp(resultsTable)
        disp('')
        writetable(resultsTable,'results.csv');
    end

    % perform the calculations
    items_of_interest();
end
```

Results

The calculations were performed by calling the function, `activity_17`. No solver issues were reported.

```
>> activity_17
          item                 value
    ----------------    ------------------

    "Final T (K)"            1235       1325
    "Time (min)"           1.4513     12.178
    "Conversion (%)"      0.79294      7.878
```

Discussion

As expected (see the preliminary analysis above), the temperature increases by 10 K during the first 1.5 min, resulting in a conversion of 0.8%. After a total of 12.2 min, the temperature has increased by 100 K and the conversion is 7.9%. Apparently this reaction is quite slow, given the very high temperatures involved.

Summary

Table 1: Required reaction time and final conversion for temperature increases of 10 and 100 K.

Temperature increase	10 K	100 K
Final Temperature (K)	1235 K	1325 K
Reaction Time (min)	1.5	12.2
Conversion	0.8%	7.9%